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PARTICULATE MATTER FILTRATION APPARATUS AND METHOD THEREOF

Abstract

A particle separator includes: a mist dispenser adapted to provide a mist; and a first fan adapted to clean an exhaust gas. The first fan includes: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member. The plurality of blades are adapted to rotate about the central chamber.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application is a divisional application of U.S. Non-Provisional application Ser. No. 17/587,205 filed Jan. 28, 2022, which claims priority to U.S. Provisional Application No. 63/227,798 filed Jul. 30, 2021, the disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] The manufacturing of semiconductor integrated circuits with an increased device density and enhanced circuit functionalities is becoming increasingly complicated. Further, the manufacturing of the integrated circuits is usually performed along with generation of polluting by-products, which may cause concerns of environmental pollution. Among the various kinds of pollutants, particulate matter (PM) is considered an important source of air pollution and has drawn a great deal of attention in recent years. PM is usually found in an exhaust gas generated during a semiconductor processing step or in an exhaust cleaning step. Although the existing exhaust PM filtration techniques have gained significant progress, there are still many aspects left to be addressed, e.g., continuing reduction of the PM mass concentration, in order to meet the more stringent air pollution control requirements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a block diagram of a semiconductor fabrication system, according to some embodiments of the present disclosure.

[0005] FIG. 2 is a block diagram of a portion of an exhaust gas cleaning apparatus, in accordance with some embodiments of the present disclosure.

[0006] FIG. 3 is a block diagram of the particle separator shown in FIG. 2, in accordance with some embodiments of the present disclosure.

[0007] FIGS. 4A and 4B show perspective views of a first fan of the particle separator shown in FIG. 2 or FIG. 3, in accordance with various embodiments of the present disclosure.

[0008] FIG. 5 shows a top view of the first fan shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure.

[0009] FIGS. 6A, 6B, 6C and 6D show perspective views of a blade of the first fan shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure.

[0010] FIGS. 7A, 7B, 7C and 7D show elevation views of a blade of the first fan shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure.

[0011] FIG. 8A shows a top view of the first fan shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure.

[0012] FIG. 8B shows a perspective view of a blade of the first fan shown in FIG. 8A, in accordance with various embodiments of the present disclosure.

[0013] FIG. 9 shows a plan view of a second fan of the particle separator shown in FIG. 2 or FIG. 3, in accordance with some embodiments.

[0014] FIG. 10 shows a block diagram of a hopper of the particle separator shown in FIG. 2 or FIG. 3, in accordance with some embodiments.

[0015] FIG. 11 shows a flowchart of a particle filtration method, in accordance with some

embodiments.

[0016] FIG. 12 is a block diagram of a computing system implementing a particle filtration method, in accordance with some embodiments.

DETAILED DESCRIPTION

[0017] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0018] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the deviation normally found in the respective testing measurements. Also, as used herein, the terms “about,” “substantial” or “substantially” generally mean within 10%, 5%, 1% or 0.5% of a given value or range. Alternatively, the terms “about,” “substantial” or “substantially” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “about,” “substantial” or “substantially.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as being from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0020] The terms “couple,” “coupled” and “coupling” used throughout the present disclosure describe the direct or indirect connections between two or more devices or elements. In some cases, a coupling between at least two devices or elements refers to mere electrical or conductive connections between them and intervening features may be present between the coupled devices and elements. In some other cases, a coupling between at least two devices or elements may involve physical contact and/or electrical connections.

[0021] The present disclosure relates generally to an exhaust cleaning system, and particularly to a particle filtration system associated with a semiconductor process exhaust. The process exhaust, or flue gas, generated during a semiconductor processing step usually contains pollutants, and thus one or more pollutant filtration processes are needed before the process gas is discharged out of the semiconductor fabrication facility. In the present disclosure, a particulate matter (PM) filtration

apparatus is introduced to aid in reducing the PM mass concentration left in the flue gas, in which the flue gas is introduced to a rotary particle separator and mixed with liquid droplets in a fluid mist. The agglomeration process of the PM particles with the mist droplets is strengthened by help of the rotary particle separator, and thus the PM particles can be captured to form enlarged PM particles. The mist-mixed PM particles therefore have a greater particle size and can be separated more easily, and thus the performance of PM filtration can thus be improved.

[0022] FIG. 1 is a block diagram of a semiconductor fabrication system **100**, according to some embodiments of the present disclosure. In the depicted example, the semiconductor fabrication system **100** is a semiconductor fabrication facility for producing semiconductor devices. In some other cases, the semiconductor fabrication system **100** can be any other kinds of facilities or equipment that generate exhaust gas or flue gas. The semiconductor fabrication system **100** may include one or more semiconductor tools **102**, one or more pumps **104**, one or more exhaust gas cleaning apparatus **106**, e.g., **106A** and **106B**, an exhaust duct **112**, a pump **114**, and an exhaust outlet **116**. One or more apparatus can be incorporated into the semiconductor fabrication system **100** but are omitted in the depicted embodiment for clarity. In some embodiments, the semiconductor fabrication system **100** includes more than one floor, and the previously mentioned equipment or devices are arranged in different floors and coupled together.

[0023] The semiconductor tool **102** is used to perform one or more semiconductor processing steps on workpieces, e.g., semiconductor wafers, for fabricating semiconductor devices. The operation type of the semiconductor processing steps include, but are not limited to, cleaning, etching, deposition, oxidation, implantation, annealing, baking, exposure, bonding, or the like. In some embodiments, during operation, one or more reacting materials or catalyst materials are introduced into the semiconductor tool **102** for performing the processing steps as desired. In some cases, exhaust gas is generated during the semiconductor processing steps. The exhaust gas of the semiconductor tool **102** is discharged to the exhaust gas cleaning apparatus **106** via the corresponding pump **104**. The exhaust gas of the semiconductor tool **102** may include pollutants that need to be filtered before being discharged to the atmosphere outside the semiconductor fabrication system **100**, e.g., PM.sub.10, PM.sub.2.5.

[0024] The pumps **104** are arranged between outlets of the semiconductor tools **102** and inlets of the corresponding exhaust gas cleaning apparatus **106**. The pumps **104** are configured to pump out the exhaust gas generated by the semiconductor tool **102** during one or more semiconductor processing steps and direct the exhaust gas into the gas cleaning apparatus **106**. In some embodiments, the pump **104** includes a dynamic type of pump, e.g., a centrifugal pump, a vertical centrifugal pump, a horizontal centrifugal pump, and the like, and a positive-displacement type pump, e.g., a diaphragm pump, a gear pump, a peristaltic pump, and the like. In some embodiments, the pump **104** includes a draft fan and include any suitable type of the draft fan, such as a centrifugal fan, an axial fan, a mixed-flow fan, or the like.

[0025] The exhaust gas cleaning apparatus **106** are configured to filter the exhaust gas generated by the semiconductor tools **102**. The exhaust gas cleaning apparatus **106** may include particle separators, which are used alone or combined with other modules according to different applications. For example, a first type exhaust gas cleaning apparatus **106A** includes a particle separator **110A** that is responsible for filtering the particles by separating the particles or other solids from the exhaust gas generated by the semiconductor tool **102A**. In some embodiments, the first type exhaust gas cleaning apparatus **106A** is more suitable where the toxic substances in the exhaust gas generated by the semiconductor tool **102A** are absent or pre-cleaned.

[0026] In another example, a second type exhaust gas cleaning apparatus **106B** is formed of a scrubber **108** and a particle separator **110B** to perform two-stage pollutant treatment on the exhaust gas. The treatment of the scrubber **108** may generate various kinds of residues after exhaust treatment, including PM of different particle sizes. The particle separator **110B** is used to capture and filter the residual particles left by the scrubber **108**.

[0027] In some embodiments, the scrubber **108** includes a wet scrubber, a semi-dry scrubber, a dry scrubber, or the like. The scrubber **108** may be used to remove or neutralize toxic or corrosive gas (e.g., acid or alkaline gas) by exposing the exhaust gas to a scrubber solution contained in the scrubber **108**. The scrubber solution may include water or other neutralizing liquids and are sprayed out to thereby cool the exhaust gas and/or adjust the pH value of the exhaust gas. In some embodiments, the scrubber solution that is used for cooling the exhaust gas in the scrubber **108** may be evaporated by the high-temperature exhaust gas, condensed with the particles and discharged. Since the scrubber **108** is focused on controlling the temperature or pH level of the exhaust gas within an acceptable range, the performance of particle filtration of the scrubber **108** may not be satisfactory. In this connection, the particle separator **110B** serve to improve the performance of particle filtration for the exhaust gas discharged by the scrubber **108**.

[0028] The exhaust gas discharged by the exhaust gas cleaning apparatus **106** flows into the exhaust duct **112**. The exhaust gases coming from different instances of semiconductor tools **102** deployed in the semiconductor fabrication system **100** are exhausted to the atmosphere through an exhaust outlet **116** and a pump **114**. The exhaust duct **112** may be formed of a main piping that is in fluid communication with a plurality of exhaust gas cleaning apparatus **106**. In alternative embodiments, the exhaust duct **112** includes a piping network formed of a main piping and branch pipes to extend its coverage across the semiconductor fabrication system **100** and to be in fluid communication with the exhaust gas cleaning apparatus **106** across different locations of the semiconductor fabrication system **100**. In some embodiments, the pumping power of the exhaust outlet **116** is greater than that of the pump **104**, e.g., the pump **114** has a pumping power of 20 to 100 times of the pumping power of each of the pumps **104**. In some embodiments, more than one pump **114** is deployed in the piping network of the exhaust duct **112** to ensure a sufficient pumping force for the discharge of the exhaust gas.

[0029] In some embodiments, the semiconductor fabrication system **100** further includes a scrubber **118** arranged between the exhaust duct **112** the exhaust outlet **116**. The scrubber **118** may be configured to perform the final stage of exhaust gas cleaning before the exhaust gas is discharged out of the semiconductor fabrication system **100**. The configuration of the scrubber **118** may be similar to the scrubber **108**; however, the processing capacity of exhaust gas of the scrubber **118** is greater than the scrubber **108** used in each of the exhaust gas cleaning apparatus **106**. By help of the scrubber **118**, the total pollutant mass concentration in the exhaust gas can be further decreased.

[0030] In some embodiments, the semiconductor fabrication system **100** further includes a plurality of sensors **122** configured to perform data sensing for the exhaust gas filtration process. The sensors **122** may include various kinds of monitors arranged at the outlets of the semiconductor tools **102** and configured to monitor the operations of the exhaust gas cleaning apparatus **106**. In some examples, the sensors **122** are configured as flow rate sensors arranged at the outlets of the semiconductor tools **102** or the pumps **104** or the exhaust outlet **116** and configured to monitor the flow rate of the fluid pumped by the pumps **104** or the pump **114**, or the exhaust gas cleaning apparatus **106**. The flow rate sensors may also be arranged at the inlets of the pumps **104** or the exhaust outlet **116**. In some other examples, the sensors **122** may include humidity sensors configured to monitor a liquid concentration of the exhaust gas or the weather humidity data in the surroundings of the semiconductor fabrication system **100**. The sensors **122** may also include various kinds of pollutant sensors arranged at the outlets of the semiconductor tools **102** and configured to detect the pollutant mass concentrations in the exhaust gas in different stages of the exhaust gas filtration process.

[0031] In some embodiments, the semiconductor fabrication system **100** further includes a processor **120** configured to receive the sensing data provided by the sensors **122**, analyze these sensing data and control the operations of the exhaust gas cleaning apparatus **106** or other the parts of the semiconductor fabrication system **100**, e.g., the rotation speed of the first fan **212** or the second fan **214**, and the flow rate of the fluid mist generated by the mist generator **206** or the mist

dispenser **306**. The processor **120** may be connected to the sensors **122** through a wired connection network or in a wireless manner. In some embodiments, the processor **120** is configured to store the sensing data in a database **124**. In some embodiments, the database **124** further stores various kinds of historical process data, e.g., from a fault detection and analysis module, that may be relied on by the processor **120** in analyzing the performance of the exhaust gas cleaning apparatus **106**, e.g., an operation type performed by the semiconductor tool **102**, a material list associated with the operations performed by the semiconductor tool **102**, an operation recipe of the operations performed by the semiconductor tool **102**, a list of pollutants generated during the operations, a mapping between operation parameters of the exhaust gas cleaning apparatus **106** and the corresponding operations, an operation recipe of the operation type, a particle mass concentration of the exhaust gas, or the like. In some embodiments, the database **124** receives and provides various kinds of real-time environment data that may be relied on by the processor **120** in analyzing the pollutant control performance, e.g., weather forecast data or air condition forecast data in an environment outside the semiconductor fabrication system **100**.

[0032] In some embodiments, the processor **120** is configured to load algorithms to conduct pollutant control analysis based on the previously mentioned sensing data, process data or environment data. The algorithms may include artificial intelligent analysis techniques, e.g., datamining techniques or an artificial neural network. The cooperation of the big data and artificial intelligence techniques may aid in increasing the analysis accuracy and stability and improve the control performance of the processor **120** to achieve an efficient, and accurate and pollutant control framework.

[0033] FIG. **2** is a block diagram of a portion of the exhaust gas cleaning apparatus **106**, in accordance with some embodiments of the present disclosure. FIG. **2** illustrates that the exhaust gas cleaning apparatus **106** includes a particle separator **110**, an exhaust gas inlet **202**, an exhaust gas outlet **204**, and a bypass duct **218**. The exhaust gas inlet **202** is in fluid communication with at least one of the semiconductor tool **102** and the scrubber **108**. The exhaust gas outlet **204** is in fluid communication with the exhaust duct **112**.

[0034] In some embodiments, the particle separator **110** includes a mist generator **206**, a motor **208**, a first fan **212**, a second fan **214** and a hopper **216**. During operation, the exhaust gas (labelled as “EG” shown in FIG. **3**) generated by the semiconductor tool **102** or the scrubber **108** is directed into the exhaust gas inlet **202**, through the particle separator **110**, e.g., the first fan **212**, the second fan **214** and the hopper **216**, and reaches the exhaust gas outlet **204**. In some embodiments, the particle separator **110** includes an input channel **222** connecting the first fan **212** to the exhaust gas inlet **202**. In some embodiments, the particle separator **110** includes an output channel **224** connecting the hopper **216** to the exhaust gas outlet **204**. The particle separator **110** may further include an inlet valve **242** arranged between the exhaust gas inlet **202** and the input channel **222** and an outlet valve **244** arranged between the exhaust gas outlet **204** and the output channel **224**, in which the inlet valve **242** and the outlet value **244** are used to switch the exhaust gas inlet **202** and the exhaust gas outlet **204**, respectively.

[0035] The mist generator **206** is configured to generate a beam of fluid mist, or simply a beam of mist, which is to be mixed with the exhaust gas in the first fan **212**. The fluid mist may facilitate the agglomeration process of the particles, e.g., the PM, in the exhaust gas and improve the performance of separating the particles from the exhaust gas. The particle separator **110** may further include a valve **207** configured to switch the dispensing of the fluid mist generated by the mist generator **206**. In some embodiments, the particle separator **110** further include drainpipes **232** at a bottom side of the first fan **212**, in which the drainpipes **232** are configured to drain the fluid condensed from the fluid mist.

[0036] The hopper **216** is configured to filter the exhaust gas at the output of the second fan **212**. The hopper **216** may include a liquid solution to facilitate capturing of liquid droplets along with PM residues left in the exhaust gas in case such solid and liquid residues are not completely filtered

by the second fan **212**, and the hopper **216** may thus improve the performance of the particle separator **110**. The particle separator **110** may further include a drainpipe **234** at a bottom side of the hopper **216**, in which the drainpipe **234** is configured to drain the liquid solution out of the hopper **216**. The drainpipes **232** and **234** may be connected to a main drainpipe **236** and the condensed liquids are drained out of the particle separator **110** through the main drainpipe **236**. [0037] The bypass duct **218** is in fluid communication with the exhaust gas inlet **202** and the exhaust gas outlet **204**. The bypass duct **218** may permit the exhaust gas to flow through when the particle separator **110** is subject to an examination work or maintenance routines. The bypass duct **218** may include a bypass inlet valve **246** and a bypass outlet valve **248** used to switch the bypass duct **218**. In some embodiments, the bypass duct **218** is not equipped with exhaust gas cleaning functions and only allows the exhaust gas to bypass the particle separator **110**. Although not illustrated, each of the bypass ducts **218** may be in fluid communication with another exhaust gas cleaning apparatus **106** or particle separator **110** of the semiconductor fabrication system **100**. As a result, the exhaust gas cleaning operation of one particle separator **110** can be taken care of by another particle separator **110**, such that the individual particle separators **110** can receive examination or maintenance work without compromising the exhaust gas cleaning performance. In some embodiments, the bypass duct **218**, the exhaust gas inlet **202** and exhaust gas outlet **204** are included in the particle separator **110**.

[0038] As illustrated in FIG. 2, the sensor **122**, the valve **207** and the motor **208** connect to the processor **120**. In some embodiments, the exhaust gas cleaning apparatus **106** further includes a controller **126** configured to control the operations of the valve **207** and the motor **208**. The controller **126** may include a programmed logic controller to control the flow rate of the fluid mist or the rotation speed of the first fan **212** or the second fan **214** through controlling the driving power of the motor **208**. In some embodiments, the exhaust gas cleaning apparatus **106** further includes a variable frequency drive configured to control the rotation speed of the motor **208** through the commands of the controller **126**.

[0039] FIG. 3 is a block diagram of the particle separator **110** shown in FIG. 2, in accordance with some embodiments of the present disclosure. In some embodiments, as illustrated in FIG. 3, the particle separator **110** includes chamber walls **300** and the particle separator **110** is compartmented by the chamber walls **300** into a first chamber **302**, a second chamber **304** and a chamber channel **305**. The exhaust gas EG is pumped into the particle separator **110** from the exhaust gas inlet **202**, through the first chamber **302**, the chamber channel **305** and the second chamber **304**, and reaches the hopper **216**, as indicated by the arrows. In some embodiments, the first fan **212** is arranged in the first chamber **302** and the second fan **214** is arranged in the second chamber **304**. In some embodiments, the second fan **214** is arranged downstream of the first fan **212**.

[0040] The particle separator **110** may further include a mist dispenser **306** connected to the mist generator **206** and extending in the exhaust gas inlet **202**. The mist dispenser **306** is configured to transport and spray the fluid mist generated by the mist generator **206**. The mist dispenser **306** also includes a dispensing line extending from the mist generator **206** to the inside of the first fan **212**. The mist dispenser **306** also includes a plurality of nozzles or sprinklers **308** on the dispensing line for dispensing the fluid mist into the input channel **222** and the first fan **212** where the exhaust gas EG is distributed. The nozzles **308** may be arranged in a row from the input channel **222** to the inside of the first fan **212**, e.g., a central location of the first fan **212**. The nozzles **308** may also be arranged to dispense the fluid mist outwardly in a radial manner from the dispensing line.

[0041] In some embodiments, the beam of fluid mist generated by the mist generator **206** includes atomized fluid droplets having a relatively small droplet size to increase the total surface area of the droplets of the fluid mist. The higher the contact area between the fluid mist and the exhaust gas EG is, the better result the agglomeration process of the particles in the exhaust gas EG may achieve. In some embodiments, the mist generator **206** is configured to generate the fluid mist of the desired droplet size by mixing the fluid with high pressure air to form the fluid mist before

sprinkling the fluid mist. Alternatively, the fluid mist is generated by utilizing high-pressure spraying to sprinkle the fluid mist out of the nozzle **308** while breaking the fluid into the mist droplets of a desirable droplet size.

[0042] In some embodiments, the mist generator **206** is configured to use water, e.g., deionized water, for generating the fluid mist. In some embodiments, the fluid mist includes other kinds of solutions, e.g., acid or alkaline solutions, to adjust, e.g., neutralize, the pH level of the exhaust gas EG.

[0043] In some embodiments, the particle separator **110** further includes a motor **228** and shafts **210** and **220**, in which the motor **228** is configured to rotate the first fan **212** through the shaft **210**, while the motor **208** is configured to rotate the second fan **214** through the shaft **220**. In some embodiments, the first fan **212** and the second fan **214** are rotated by a same motor, e.g., the motor **208**. In some embodiments, the shafts **210** and **220** rotate about the same or different axes.

[0044] As illustrated in FIG. 3, the first fan **212** includes an inner mesh member **312**, a plurality of blades **314**, an outer mesh member **316**, an upper cover **406** and a lower cover **408**. FIGS. 4A and 4B show perspective views of the first fan **212** in FIG. 2 or FIG. 3, in accordance with various embodiments of the present disclosure. The first fan **212** can be referred to herein as a particle separator fan, while the second fan **214** can be referred to herein as a draft fan, in which the first fan **212** serves to capture the particles in the exhaust gas EG and filter these particles through mixing with the fluid mist, while the second fan **214** serves to generate an airflow to draw the exhaust gas EG from the exhaust gas inlet **202** into the first fan **212** during a particle filtration operation and pump the cleaned exhaust gas EG out of exhaust gas outlet **204**.

[0045] The upper cover **406** and the lower cover **408** are arranged on an upper side and a lower side, respectively, of the first fan **212**. The upper cover **406** and the lower cover **408** are attached to the inner mesh member **312**, the blades **314** and the outer mesh member **316** to form an enclosed space of the first fan **212** from the upper side and the lower side of the first fan **212**. In some embodiments, the parts of the first fan **212**, e.g., the inner mesh member **312**, the blades **314**, the outer mesh member **316**, the upper cover **406** and the lower cover **408** are formed of rigid materials, e.g., plastic, polymer, metal, stainless steel, or the like. These parts may be multilayer structures or include a coating formed thereon.

[0046] In some embodiments, the first fan **212** has a cylinder shape. The inner mesh member **312** is in a ring shape or a polygonal shape when viewed from above. The inner mesh member **312**, the upper cover **406** and the lower cover **408** define a central chamber **318** at a central location of the first fan **212**. Similarly, the outer mesh member **316** is in a ring shape or a polygonal shape when viewed from above and defines the periphery of the first fan **212**. Each of the blades **314** includes one end coupled to the sidewall of the inner mesh member **312** and the other end coupled to the sidewall of the outer mesh member **316**. The lower cover **408** may include an opening **408A** facing the input channel **222** and connecting the input channel **222** to the central chamber **318**, wherein the mist dispenser **306** extends into the central chamber **318** through the opening **408A**. In this connection, the exhaust gas EG can flow into the first chamber **302** or the first fan **212** only through the opening **408A**.

[0047] FIG. 5 shows a top view of the first fan **212** shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure. Some parts of the first fan **212**, e.g., the upper cover **406**, are omitted from FIG. 5 in order not to obscure the underlying features of the particle separator **110**. Referring to FIGS. 3, 4A, 4B and 5, at the beginning stage of operation, the first fan **212** and the second fan **214** are rotated through the motors **228** and **208**, respectively. In the meantime, the fluid mist is dispensed to the central chamber **318** of the second fan **212**. The fluid mist is diffused by the air pressure exerted by the first fan **212** and the second fan **214** from the central chamber **318** to the space of the blades **314** between the inner mesh member **312** and the outer mesh member **316**. The blades **314** laterally surround the central chamber **318** and configured to rotate about the central chamber **318**.

[0048] FIG. 6A shows a perspective view of an exemplary blade **314A** of the first fan **212** shown in FIG. 4A and FIG. 4B, in accordance with various embodiments of the present disclosure. The blade **314A** has a curve shape when viewed from above. The blade **314A** has a substantially equal thickness throughout the wall of the entire blade **314A**. The blade **314A** includes holes **314X** arranged in an array on the wall of the blade **314A**. In the present embodiment, the holes **314X** are arranged in an array of five columns and 9 rows; however, the configurations of the holes **314X** may be different, as illustrated in other figures discussed later.

[0049] Referring to FIGS. 4A and 4B, each of the inner mesh member **312** and the outer mesh member **316** include a mesh pattern that forms a plurality of holes (an example hole **316H** is labelled in FIG. 4A) to permit the exhaust gas EG along with the fluid mist to flow through when the exhaust gas EG is pumped into the space between the blades **314** from the central chamber **318** or to flow out of the outer mesh member **316**. These holes **316H** may be arranged in rows, in columns, or in an array on the walls of the inner mesh member **312** and the outer mesh member **316**. Alternatively, the inner mesh member **312** or the outer mesh member **316** includes the holes **316H** randomly distributed on the wall thereof.

[0050] The mesh pattern or the configuration of the holes **316H** of the inner mesh member **312** and the outer mesh member **316** is determined to cause liquid membranes to be formed on the holes **316H** to serve as a particle catcher for increasing the probability of particle filtration. In some embodiments, the holes **316H** have a circular shape, a polygonal shape, a crescentic shape, or any suitable shape. The size of the holes **316H** may be determined based on factors such as the flow rate of the exhaust gas EG, the droplet size of the fluid mist, and the like. In the present embodiment, the holes **316H** have a size in a range between about 0.5 mm and about 3 mm. In some embodiments, the holes **316H** have a size in a range between about 1 mm and about 2 mm. If the hole size is made less than about 0.5 mm, the resistance to air flow, which results from the wall of the inner mesh member **312** or the outer mesh member **316**, may be too high and hinder the flow of the exhaust gas EG. If the hole size is made greater than about 3 mm, the liquid membrane may not be formed easily. The configurations of the mesh pattern or the holes **316H**, e.g., the number, the density, the shape, and the array pattern, may be varying depending upon different particle filtration requirements and are within the contemplated scope of the present disclosure.

[0051] In some embodiments, the inner mesh member **312** is omitted, and the blades **314** are secured to the upper cover **406**, the lower cover **408** and the outer mesh member **316**. In such scenarios, the fluid mist is directly drawn into the space between the blades **314** with a greater flow rate, thereby causing the fluid mist to be uniformly distributed across the first fan **212** more quickly.

[0052] The exhaust gas EG and the particles are carried through the first fan **212** in two types of movements. In the first type movement, the airflow generated by of the second fan **214** draws the exhaust gas EG and the particles out of the outer mesh member **316**. The exhaust gas EG moves in a radial or straight-line trajectory in the first type movement. In the second type movement, the blades **314** rotate the exhaust gas EG and the particles in the space between the blades **314**. As illustrated in FIG. 5, the second fan **212** is rotated in a counterclockwise direction **D1**, and thus an example particle **P1** in the exhaust gas EG is rotated by the blades **314** in the counterclockwise direction **D1**. As a result, the combination of the first and second movements causes the exhaust gas EG and the particle **P1** to move in a spiral-like trajectory when viewed from above until they flows out of the first fan **212**.

[0053] In the meantime, the fluid mist is dispensed to fill the space between the blades **314** by the pumping force of the second fan **214** and the rotation force of the blades **314**, in a similar trajectory of the exhaust gas EG and the particle **P1**. The agglomeration process occurs when the fluid mist is mixed with the particle **P1**. In some embodiments, homogeneous nucleation of the particle **P1** with the droplets of the fluid mist facilitates the agglomeration process, which causes the effective size of the fluid-mixed particle **P1** in the exhaust gas EG to grow, e.g., from less than about 0.1 μm to

greater than about 5 μm , along the movement trajectory. Referring to FIG. 3, the droplets of the fluid mist along with the enlarged particle **P1** flowing out of the outer mesh member **316** may condense and be drained out of the first fan **212** through the drainpipes **232**. The particle **P1** is thus separated from the exhaust gas EG due to its greater particle size and increased particle mass, and the cleaned exhaust gas EG can be obtained it flows out of the first fan **212**.

[0054] The performance of particle separation is determined by the likelihood of mixing of the particles with the droplets through the agglomeration process. Thus, the particle separator **110** provides advantageous aspects to increase the likelihood of the agglomeration process, such as increasing the total area of the droplets of the fluid mist, increasing the length of the movement trajectory of the particles in the particle separator **110**, increasing the distribution uniformity of the fluid mist in the particle separator **110**, or the like.

[0055] It can be observed that the spiral trajectory of the movement for the particle **P1** has a greater length than the radial trajectory caused by a conventional particle separator which does not employ the rotary blades **314**. Since the movement trajectory of the particle **P1** is lengthened, the duration of stay of the particle **P1** in the first fan **212** is increased and the probability of agglomeration of the particle **P1** is increased accordingly. As a result, the filtration performance of the particle separator **110** is improved.

[0056] The droplet size of the fluid mist is one of the factors that influence the performance of the agglomeration process. The fluid mist with smaller droplets may lead to greater surface areas of the droplets, and hence increases the probability of contact with the particles. In some embodiments, the droplet size of the fluid mist is less than about 200 μm , less than about 100 μm or less than about 50 μm . In some embodiments, droplet size of the fluid mist is in a range between about 20 μm and about 100 μm , between about 20 μm and about 50 μm or between about 50 μm and about 100 μm . If the droplet size is made greater than about 200 μm , the droplets may not have sufficient contact areas to effectively capture the particles. If the droplet size is made less than about 20 μm , the resultant agglomerated particles may not be large enough to be separated from the exhaust gas EG and drained efficiently.

[0057] In some embodiments, the performance of the agglomeration process is also close related to the flow rate of the fluid mist as well as the flow rate of the exhaust gas EG. In some embodiments, the mass flow ratio between the fluid mist and the exhaust gas is in a range between about 2 and about 5. As a result, the rotation speed of the first fan **212** or the second fan **214** is determined according to the chosen flow rate of the exhaust gas EG. In some embodiments, the first fan **212** or the second fan **214** is rotated at a rotation speed between about 600 rpm and 6000 rpm, or between about 900 rpm and about 4000 rpm. In some embodiments, the first fan **212** or the second fan **214** provides a flow rate between about 500 cubic meters per hour (cmh) and 3500 cmh, or between about 1500 cmh and about 2500 cmh. In some embodiments, the dispensing rate of the mist generator **206** is between about 5 liters per minute (lpm) and about 10 lpm.

[0058] In some embodiments, the configuration of the holes **314X** may affect the agglomeration process. The holes **314X** may have a size in a range between about 0.5 mm and about 3 mm, or between about 1 mm and about 2 mm. If the holes **314X** are made greater than about 3 mm, the blades **314** may not effectively carry the exhaust gas EG to rotate in the first fan **212**. As a result, the movement trajectory of the particles may not achieve the maximal length. If the holes **314X** are made less than about 0.5 mm, it is not easy for the droplets of the fluid mist to flow in different compartments between the blades through the holes **314X**. As a result, most of the droplets are confined in one or two compartments between the adjacent blades **314**. The fluid mist of small droplets staying in the limited space of the compartment may tend to combine into large droplets, and the total contact area of the fluid mist may decrease accordingly. The blades **314** with the holes **314X** not only serves the function of rotate the exhaust gas EG to increase the movement trajectory, but also agitates the fluid mist to such that the tiny mist droplets can be distributed in the first fan **212**. As such, the holes **314X** with a sufficient hole size, e.g., greater than about 0.5 mm, may

facilitate distribution of the fluid mist across different compartments across the entire first fan **212**, and decrease the chances of recombination of the droplets in the first fan **212** before the agglomeration process.

[0059] FIGS. **6B**, **6C** and **6D** show perspective views of exemplary blade **314B**, **314C** and **314D**, respectively, of the first fan **212** shown in FIG. **4A** and FIG. **4B**, in accordance with various embodiments of the present disclosure. The blades **314B** through **314D** are similar to the blade **314A** in many aspects, and these similar features are not repeated for brevity. Referring to FIG. **6B**, the blade **314B** has the wall with a planar or flat surface from a cross-sectional view. Referring to FIG. **6C**, the blade **314C** has the wall with a doubly curved surface or a wavy surface from a cross-sectional view. Referring to FIG. **6D**, the blade **314D** has the wall with an airfoil shape from a cross-section view. The different shapes of the walls of the blades **314A** through **314D** may generate different types of fluid distributions for the exhaust gas EG as well as the fluid mist. The selection of suitable types of the blades **314** may be based on different applications of particle filtration. Although not explicitly illustrated, other shapes of the walls of the blades **314** are also within the contemplated scope of the present disclosure.

[0060] FIGS. **7A**, **7B**, **7C** and **7D** show elevation views of blade **314E**, **314F**, **314G** and **314H**, respectively, of the first fan **212** shown in FIG. **4A** and FIG. **4B**, in accordance with various embodiments of the present disclosure. The blades **314E** through **314H** shown in respective FIGS. **7A** through **7D** can be drawn from any of FIGS. **6A** through **6D**. Referring to FIG. **7A**, the holes **314X** of the blade **314E** are arranged with substantially equal column pitches **T1** and substantially equal row pitches **T2**. In some other embodiments, the column pitch **T1** and the row pitch **T2** may vary across different columns or rows. For example, referring to FIG. **7B**, the holes **314X** of the blade **314F** are arranged with different column pitches **T1** and substantially equal row pitches **T2**. The different column pitches **T1** for different columns may help redistribute the fluid mist more quickly and uniformly across the first fan **212**, e.g., from the end adjacent to the central chamber **318** to the end adjacent to the outer mesh member **316**.

[0061] Referring to FIG. **7C**, although the column pitch **T1** and the row pitch **T2** of the holes **314X** of the blade **314F** are substantially the same across different columns and rows, the holes **314X** may be arranged in a staggering manner across different columns. Referring to FIG. **7D**, the layout of the holes **314X** of the blade **314H** is further altered such that the numbers of the holes **314X** may be different across different columns or rows. In the present embodiments, the number of holes **314X** are monotonically decreasing from the end adjacent to the central chamber **318** to the end adjacent to the outer mesh member **316**. However, other configurations of the numbers of the holes **314X** are also within the contemplated scope of the present disclosure.

[0062] FIG. **8A** shows a top view of a first fan **252** shown in FIG. **4A** and FIG. **4B**, in accordance with various embodiments of the present disclosure. The first fan **252** is similar to the first fan **212** shown in FIG. **3** in many aspects, such as the inner mesh member **312**, the outer mesh member **316** and the central chamber **318** discussed previously, and the descriptions of these features are omitted for brevity. The first fan **252** also includes blades **324** arranged in the space between the inner mesh member **312** and the outer mesh member **316** in a manner similar to the blades **314**. The blade **324** differs from the blade **314** in that the blade **324** is formed of individual scramblers **326** extending between the upper cover **406** and the lower cover **408**, in which the scramblers **326** are separate from each other and are arranged in parallel.

[0063] FIG. **8B** shows a perspective view of an example blade **324** of the first fan **252** shown in FIG. **8A**, in accordance with various embodiments of the present disclosure. As illustrated in FIGS. **8A** and **8B**, the scramblers **326** of each of the blades **314** are arranged in a curved line when viewed from above. In some embodiments, each of the scramblers **326** includes a cylinder, a cuboid, a polygonal prism, or the like. In some embodiments, the scrambler **326** includes a circular shape, a polygonal shape or a ring shape from a top-view perspective. The blade **324** has a scrambler spacing **S1** between the adjacent scramblers **326**. In some embodiments, the scrambler spacing **S1**

is substantially uniform from the bottom of the blade **324** to the top of the blade **324**. In some embodiments, the scrambler spacing **S1** is between about 0.5 mm and about 3 mm. In some embodiments, the scrambler spacing **S1** between the scramblers **326** is seen as an elongated hole **314X** of the blade **324**, in which the hole **314X** extends from the bottom of the blade **324** to the top of the blade **324**. The design of the scrambler **326** with the scrambler spacing **S1** of the blade **324** may increase the hole area of the blade while decreasing the manufacturing cost of the blade **324**. The width of the hole **314X** may be kept as a predetermined size of the hole **314X** shown in FIGS. **6A** through **6D** and **7A** through **7D**. In some embodiments, the liquid membrane formed from the liquid mist on the hole **314X** has a greater area as compared to that formed on the hole **314X** of the blade **314**. The particle filtration performance may be enhanced accordingly.

[0064] The different types of the blades **314** with different arrangements of the walls and holes **314X** as shown in FIGS. **6A** through **6D**, FIGS. **7A** through **7D** and FIGS. **8A** and **8B** may generate different types of fluid distributions for the exhaust gas **EG** as well as the fluid mist during the agglomeration process. The selection of suitable types of the blades **314** may be based on different applications of particle filtration. Although not explicitly illustrated, other types of the blades **314** are also within the contemplated scope of the present disclosure.

[0065] Referring to FIG. **3**, as discussed previously, the second fan **214** is configured as a draft fan to exert a pumping force on the exhaust gas **EG**. The second fan **214** can be any suitable type of draft fan, such as a centrifugal fan, an axial fan, a mixed-flow fan, or the like. In some embodiments, the centrifugal fan may be categorized as a forward curved blade type, a backward curved blade type and a radial blade type. The operation parameters of the second fan **214**, such as the rotation speed, air pressure, air flow rate, supply current levels, are determined according to different requirements of the semiconductor fabrication system **100** and the conditions of the semiconductor tools **102**.

[0066] FIG. **9** shows a plan view of an example second fan **214** of the particle separator **110** shown in FIG. **2** or FIG. **3**, in accordance with some embodiments. The second fan **214** illustrated in FIG. **9** is a centrifugal fan and includes a volute casing **502**, an impeller **506** and a shaft **508**. Referring to FIG. **2** and FIG. **9**, the volute casing **502** defines a volute chamber **504** that encloses the impeller **506**, where the shaft **508** extends from the motor **208**, through the volute casing **502** and connects to the impeller **506**. The impeller **506** is rotated by the motor **208** through the rotation of the shaft **508**. The other portions of the volute chamber **504** serve as an air channel for the flow of the exhaust gas **EG**, and the exhaust gas **EG** flows along the path the spiral shape of the volute casing **502** to reach the hopper **216**.

[0067] An opening **514** is formed on the volute casing **502** and corresponding to the center of the impeller **506** around the shaft **508**. The opening **514** is arranged facing the chamber channel **305** and the first fan **212**, where the cleaned exhaust gas **EG** exhausted from the first fan **212** is directed into the second fan **214** from the opening **514**. The opening **514** may be in an annular shape or other suitable shape and laterally surround the shaft **508**.

[0068] In some embodiments, the impeller **506** includes a plurality of vanes or blades **512** connected to the shaft **508** and radially extended from the shaft **508**. In some embodiments, the impeller **506** may be of different configurations, such as an open-type impeller, a semi open-type impeller, and an enclosed-type impeller. In the embodiment of the impeller **506** of a semi-open type or an enclosed type, a shroud or disk (not separately shown) is provided on one side or two sides of the blades **512** in a plan substantially perpendicular to the shaft, in which the adjacent blades **512** are fully or partially adjoined by the shroud. The impeller type of the impeller **506** is selected according to the requirements of the particle separator **110**.

[0069] In some embodiments, the performance of particle filtration is closely related to factors such as the radial flow speed of the exhaust gas **EG** in the first fan **212**, the rotation speed of the first fan **212**, the (molecular or atomic) weight of the particles to be filtered, the mass concentration of the particles to be filtered, and the like. In some embodiments, the rotation speed of the first fan **212** or

the second fan **214** is inversely proportional to the weight of the particles to be filtered in order to provide substantial equal centrifugal forces for different types of particles.

[0070] In some embodiments, the rotation speed of the second fan **214** is in a range between about 1,800 rpm and about 4,200 rpm. In some embodiments, the flow rate of the second fan **214** is in a range between about 1,000 lpm about 6,000 lpm.

[0071] FIG. **10** shows a block diagram of the hopper **216** of the particle separator **110** shown in FIG. **2** or FIG. **3**, in accordance with some embodiments. The hopper **216** includes a hopper wall **602** having a U-shape channel configured to direct the exhaust gas EG discharged by the second fan **214** toward the exhaust gas outlet **204**. In some embodiments, the channel of the hopper **216** includes a bottom portion **602B** at the bottom of the hopper **216**. The bottom portion **602B** may contain liquid, e.g., water or other suitable solution, configured to absorb the residual fluid mist and the particles left in the exhaust gas EG passing through the hopper **216**. During operation, the exhaust gas EG along with the residual fluid mist is exhausted out of the particle separator **110** by the second fan **214**. The centrifugal force exerted on the fluid mist may further drive the fluid mist to move outwardly when the exhaust gas EG flows through the bottom portion **602B**. As a result, the fluid mist along with the particles may fall into the liquid at a corner of the bottom portion **602B**, and the performance of particle filtration and dehydration of the particle separator **110** can be improved by the hopper **216**.

[0072] FIG. **11** shows a flowchart of a particle filtration method **700**, in accordance with some embodiments. It shall be understood that additional steps can be provided before, during, and after the steps shown in FIG. **11**, and some of the steps described below can be replaced or eliminated, for additional embodiments of the method **700**. The order of the steps may be interchangeable. Some of the steps may be performed concurrently or independently.

[0073] At step **702** a workpiece is processed in a semiconductor apparatus. The processing of the workpiece causes generation of an exhaust gas. At step **704**, the exhaust gas is exhausted from the semiconductor apparatus to a particle separator. In some embodiments, the particle separator includes a first fan having a central chamber and a plurality of blades.

[0074] At step **706**, a mist is dispensed into the central chamber to mix with the exhaust gas. At step **708**, the first fan is rotated while the mist is dispensed to permit mixing of the exhaust gas with the mist between the central chamber and a periphery of the particle separator.

[0075] FIG. **12** is a block diagram of a computing system **800** implementing a particle filtration method, in accordance with some embodiments. The system **800** may be operated together with a semiconductor fabrication system (e.g., the semiconductor fabrication system **100**) or a particle separator (e.g., the exhaust gas cleaning apparatus **106**) alone. The system **800** includes a processor **801**, a network interface **803**, an input and output (I/O) device **805**, a storage **807**, a memory **809**, a bus **808**, a sensor **811** and a controller **813**. The bus **808** couples the network interface **803**, the I/O device **805**, the storage **807**, the memory **809**, the sensor **811**, the controller **813** and the processor **801** to each other.

[0076] The processor **801** may correspond to or be included in the processor **120**. The processor **801** is configured to execute program instructions configured to perform the steps of particle filtration or exhaust gas cleaning as described and illustrated with reference to figures of the present disclosure. In some embodiments, the processor **801** is configured to control sensing, acquisition and analysis of the sensing data provided by the sensor **811**. In some embodiments, the processor **801** is configured to control the operations of the semiconductor fabrication system **100** for performing exhaust gas cleaning through the controller **813**. The controller **813** may be similar to the controller **126**.

[0077] The network interface **803** is configured to access program instructions and data accessed by the program instructions stored remotely through a network (not shown). In some embodiments, the network interface **803** connects the processor **801** to the parts of the lithography apparatus. In some embodiments, the network interface **803** connects the lithography apparatus to external

devices.

[0078] The I/O device **805** includes an input device and an output device configured for enabling user interaction with the system **800**. In some embodiments, the input device comprises, for example, a keyboard, a mouse, a trackball, a touchpad, a fingerprint sensor, and other devices. Moreover, the output device comprises, for example, a display, a printer, and other devices.

[0079] The storage device **807** is configured for storing program instructions and data, e.g., the recipe data, the pollutant list data, the exhaust sensing data or weather forecast data, accessed by the program instructions. In some embodiments, the storage device **807** comprises a non-transitory computer readable storage medium, for example, a magnetic disk and an optical disk.

[0080] The memory **809** is configured to store program instructions to be executed by the processor **801** and data accessed by the program instructions. In some embodiments, the memory **809** comprises any combination of a random-access memory (RAM), some other volatile storage device, a read only memory (ROM), and some other non-volatile storage device.

[0081] Some embodiments of the present disclosure provide a particle separator. The particle separator includes: a mist dispenser adapted to provide a mist; and a first fan adapted to clean an exhaust gas. The first fan includes: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member. The plurality of blades are adapted to rotate about the central chamber.

[0082] Some embodiments of the present disclosure provide a particle separator. The particle separator includes: a mist generator adapted to generate a mist; and a first fan adapted to capture particles in an exhaust gas. The first fan includes: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member and coupled to the inner mesh member.

[0083] Some embodiments of the present disclosure provide a particle separator. The particle separator includes a first fan adapted to capture particles in an exhaust gas; and a second fan arranged downstream of the first fan and adapted to draw the exhaust gas cleaned via the first fan out of the particle separator. The first fan includes: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member. The plurality of blades are adapted to rotate about the central chamber.

[0084] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A particle separator, comprising: a mist dispenser adapted to provide a mist; and a first fan adapted to clean an exhaust gas and comprising: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member, wherein the plurality of blades are adapted to rotate about the central chamber.
2. The particle separator of claim 1, further comprising a second fan adapted to generate an airflow to draw the exhaust gas and the mist out of the particle separator.

3. The particle separator of claim 2, wherein the second fan is arranged downstream of the first fan.
 4. The particle separator of claim 1, wherein the inner mesh member comprises holes, and the mist is adapted to flow into spaces between the plurality of blades through the holes of the inner mesh member.
 5. The particle separator of claim 1, wherein the mist dispenser extends into the central chamber.
 6. The particle separator of claim 1, further comprising an upper cover and a lower cover on two sides of the particle separator, wherein the lower cover defines an opening that permits the exhaust gas to flow into the particle separator.
 7. The particle separator of claim 1, wherein the outer mesh member is arranged near a periphery of the particle separator and laterally surrounding the plurality of blades.
 8. The particle separator of claim 1, wherein the outer mesh member comprises holes, wherein the exhaust gas is adapted to flow out of the particle separator through the holes of the outer mesh member.
 9. The particle separator of claim 1, wherein the particle separator further comprises a pipe adapted to drain a liquid condensed from the mist in the first fan.
 10. The particle separator of claim 1, further comprising a chamber accommodating the first fan and the mist dispenser.
 11. The particle separator of claim 1, wherein at least one of the blades comprises holes allowing the exhaust gas to pass through, and the holes have a size between about 0.5 mm and about 3 mm.
 12. The particle separator of claim 1, wherein the mist has a droplet size in a range between about 20 μm and about 100 μm .
 13. A particle separator, comprising: a mist generator adapted to generate a mist; and a first fan adapted to capture particles in an exhaust gas and comprising: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member and coupled to the inner mesh member.
 14. The particle separator of claim 13, further comprising an upper cover and a lower cover on two sides of the particle separator, wherein each of the plurality of blades comprises a plurality of scramblers arranged in parallel and extending from the upper cover to the lower cover.
 15. The particle separator of claim 14, wherein each of the scramblers are arranged in a curved line from a top-view perspective.
 16. The particle separator of claim 13, further comprising a second fan adapted to generate an airflow to draw the exhaust gas out of the particle separator.
 17. The particle separator of claim 13, wherein at least one of the plurality of blades comprises holes arranged in a staggered manner.
 18. A particle separator, comprising: a first fan adapted to capture particles in an exhaust gas and comprising: an outer mesh member; an inner mesh member laterally surrounded by the outer mesh member and defining a central chamber of the particle separator; and a plurality of blades arranged between the outer mesh member and the inner mesh member, wherein the plurality of blades are adapted to rotate about the central chamber; and a second fan arranged downstream of the first fan and adapted to draw the exhaust gas cleaned by the first fan out of the particle separator.
 19. The particle separator of claim 18, further comprising an input channel in communication with a semiconductor tool, the input channel adapted to guide the exhaust gas generated by the semiconductor tool into the first fan.
 20. The particle separator of claim 18, further comprising a chamber channel arranged between the first fan and the second fan, and allowing the exhaust gas from the first fan to the second fan, the chamber channel overlapping an axis of the first fan from a top-view perspective.
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